A03: Independent mathematical audit of P03
Origin and homogeneous linearization

Independent reconstruction and adversarial review

27 September 2026

1 Main verdict and exact version

MAIN VERDICT: VERIFIED. Every substantive mathematical assertion in the exact P03 version identified below passes independent reconstruction under its stated canonical assumptions. There is no required mathematical or wording correction. The optional nonzero-equilibrium simplification also passes within its stated domain; it needs no separate verdict.

The audited file is `proofs/P03_origin_and_linearization.tex`, **P03 v1.0, 2026-09-27**, exactly **18,921 bytes and 433 lines**, from `cyclic_network_project_P03_draft.zip`. Its full-file SHA-256, including comments, whitespace and final newline, is

`0f87f315817429ddb949af5bc6d38db32a0cb6a8f7715dfe941840becd527776`

The supplied ZIP's SHA-256 is

`768c76b7b53e0af2807d3eab81cbe38134425c3ff477585559428b9614d1fb96`

All P03 line references below use that exact source, including comment and blank lines. The hash agrees with the manifest's final delivered-component record, not its earlier pre-P03 input inventory. The supplied archive is the version authority for this audit. No placeholder or archived blueprint is used as the proof being audited.

Other local copies were checked for version confusion: the only substantive P03 draft found has the same hash. Older 1,577-byte TODO placeholders have hash beginning `8906056fbc57d1de` and were excluded.

The verdict rests on the analytic proofs in this document. Compilation, finite samples and the earlier draft's validation report are not proof premises. P03 and all supplied project files remain unchanged. A03 records that this exact version passes; the unchanged P03 header still records its pre-audit DRAFT state. Recording administrative audit closure is distinct from altering the audited proof. P04 is cleared after that closure is recorded; P04 is not begun here.

2 Current inputs, assumptions and conventions

All thirteen requested current inputs were located in the supplied archive before reconstruction. The accompanying `input_hashes.json` records their full SHA-256 digests and sizes.

Current file	Use in this audit
`project_manifest.md`	Final P03 version/hash; P01/P02 closure; current status precedence.
`claim_registry.md`	C034, C035, C037 and the P03 overlay; distinction between archived claims and current statements.
`frozen_theorem_package.md`	S1 supporting role and the P03 then P04 attempt order.
`final_novelty_verdict.md`	Frozen literature scope and the separate, unclosed priority gate.
`02_theorem_dependency_map.tex`	P01/P02 to P03 to P04 dependency graph.
`00_canonical_model.tex`	Original vector field, parameter set, modular indexing and open domain.
`01_notation_and_assumptions.tex`	Predecessor shift, DFT sign, norms and hyperbolicity definition.
`proofs/P01_wellposedness_and_absorbing_box.tex`	Current v1.1 smoothness on the open domain.
`audits/A01_wellposedness_and_absorbing_box.tex`	Original audit and sole E01 correction, now closed.
`proofs/P02_synchronous_subspace_and_equilibria.tex`	Current v1.1 scalar restriction and nonzero-equilibrium relation.
`audits/A02_synchronous_subspace_and_equilibria.tex`	Restored original audit; sole E01 correction, now closed.
`proofs/P03_origin_and_linearization.tex`	Exact 433-line mathematical target.
`proofs_compendium.tex`	Current status banner, scope and unchanged module inclusion order.

The field and the full standing assumptions are

$$F_i(x) = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+q}} - hx_ix_{i-q}, \tag{1}$$

$$n \in \mathbb{Z}, \quad n \geq 5, \quad 1 \leq q \leq n-1, \quad a,k,c,g,h > 0, \quad x \in (-1,\infty)^n. \tag{2}$$

Indices lie in $\mathbb{Z}/n\mathbb{Z}$ with representatives $0,\dots,n-1$. There is no coprimality assumption. Neighbor occurrences are retained even when their coordinates coincide. In particular, the audit includes negative coordinates in the open domain; it does not assume that this whole domain is forward invariant. The present linear statements do not need such invariance.

Write $A = k + h$ and $\mu = a + c - g$. Exactly the project's conventions are used:

$$(Pz)_i = z_{i-1}, \qquad (P^r z)_i = z_{i-r}, \qquad \theta_m = \frac{2\pi m}{n}, \qquad v_j^{(m)} = n^{-1/2} e^{\mathrm{i}\theta_m j}.$$

The Fourier vectors have a positive exponent, so the predecessor shift has a negative exponent in its eigenvalue. Hyperbolicity means that no eigenvalue has zero *real part*. All stability equivalences below concern the autonomous linear equation $\dot{z} = J(0)z$ in finite-dimensional real space.

3 Independent general Jacobian derivation

Let $\delta_{j\ell}$ denote modular equality. Differentiate the five terms of (1) separately before merging any entries:

$$\begin{aligned}
\partial_j(ax_i) &= a\delta_{ji},\\
\partial_j(-kx_i^2) &= -2kx_i\delta_{ji},\\
\partial_j\left(\frac{cx_{i-1}}{1+x_i}\right) &= \frac{c}{1+x_i}\delta_{j,i-1} - \frac{cx_{i-1}}{(1+x_i)^2}\delta_{ji},\\
\partial_j\left(-\frac{gx_i}{1+x_{i+q}}\right) &= -\frac{g}{1+x_{i+q}}\delta_{ji} + \frac{gx_i}{(1+x_{i+q})^2}\delta_{j,i+q},\\
\partial_j(-hx_ix_{i-q}) &= -hx_{i-q}\delta_{ji} - hx_i\delta_{j,i-q}.
\end{aligned}$$

The denominator derivative of the positive input has an algebraic minus sign; the denominator derivative of the negative loss has an algebraic plus sign. Their numerical signs need not equal their algebraic signs when the numerator coordinate is negative. All denominators here are nonzero.

Summing gives precisely

$$\begin{aligned}
J_{ij}(x) = &\left(a - 2kx_i - \frac{cx_{i-1}}{(1+x_i)^2} - \frac{g}{1+x_{i+q}} - hx_{i-q}\right)\delta_{ji}\\
&+ \frac{c}{1+x_i}\delta_{j,i-1} + \frac{gx_i}{(1+x_{i+q})^2}\delta_{j,i+q} - hx_i\delta_{j,i-q}.
\end{aligned} \tag{3}$$

This verifies P03 **L33–70**. A modular Kronecker symbol is an indicator appearing in a sum, not an instruction to overwrite an entry. Differentiation with respect to a repeated coordinate differentiates every occurrence of it. No application of a distinct-neighbor assumption is hidden in the derivation.

For an independent directional reconstruction, at fixed x and z use $(u+tv)^{-1} = u^{-1} - tv/u^2 + O(t^2)$ with $u > 0$. The rational first-order terms are

$$c\left(\frac{z_{i-1}}{1+x_i} - \frac{x_{i-1}z_i}{(1+x_i)^2}\right), \qquad -\frac{gz_i}{1+x_{i+q}} + \frac{gx_iz_{i+q}}{(1+x_{i+q})^2}.$$

The other terms contribute $(a-2kx_i)z_i$ and $-h(x_{i-q}z_i + x_iz_{i-q})$. This independently reproduces every coefficient of (3), including at colliding offsets.

4 Shift orientation and exhaustive local collisions

Define D_i, C_i, G_i, H_i by the four coefficients in (3), with $H_i = hx_i$ so its contribution is $-H_i$. The directional derivative is

$$(DF(x)z)_i = D_iz_i + C_iz_{i-1} + G_iz_{i+q} - H_iz_{i-q}.$$

From the indices themselves, $z_{i-1} = (Pz)_i$, $z_{i+q} = (P^{-q}z)_i$, and $z_{i-q} = (P^qz)_i$. Therefore

$$J(x) = \operatorname{diag}(D_i) + \operatorname{diag}(C_i)P + \operatorname{diag}(G_i)P^{-q} - \operatorname{diag}(H_i)P^q. \tag{4}$$

The diagonal matrices multiply on the left because the coefficient belongs to receiving row i. Right multiplication would generally sample a different row's coefficient. This verifies P03 **L73–92** without deducing orientation from its claimed answer. A nonsynchronous Jacobian need not be circulant.

No neighbor can equal i: $1 \not\equiv 0$ and $q \not\equiv 0$ modulo n. There are exactly three unordered pairs of neighbor occurrences to examine:

$$\begin{aligned} i-1 \equiv i-q &\iff q=1, \\ i-1 \equiv i+q &\iff q=n-1, \\ i+q \equiv i-q &\iff n \mid 2q \iff n \text{ even and } q=n/2. \end{aligned}$$

For the last equivalence, $0 < 2q < 2n$ forces $2q = n$. The three values $1, n-1, n/2$ cannot coincide for $n \geq 5$; any triple collision would imply two such cases and is impossible. These pairwise comparisons exhaust the list. A nontrivial $\gcd(n, q)$ affects repeated step-q paths, not these one-row congruences, so it creates no additional local collision.

The actual general-state merged entries are

$$\begin{aligned} q=1,\ j=i-1: &\quad \frac{c}{1+x_i} - hx_i, \\ q=n-1,\ j=i-1: &\quad \frac{c}{1+x_i} + \frac{gx_i}{(1+x_{i-1})^2}, \\ n \text{ even},\ q=n/2,\ j=i+q=i-q: &\quad \frac{gx_i}{(1+x_{i+q})^2} - hx_i. \end{aligned}$$

In each case the remaining neighbor contribution is retained. A merged entry may cancel to zero. This does not change the differentiation rule.

5 Origin derivative, complete spectrum and multiplicities

All five terms vanish at $x = 0$, so $F(0) = 0$. In the derivative:

- $-2kx_i$ and both h-term derivatives vanish because they contain a state coordinate.
- The c denominator derivative vanishes because it contains x_{i-1}.
- The g denominator derivative vanishes because it contains x_i.
- The numerator derivatives survive as c at $i-1$ and $-g$ at i; the linear term contributes a at i.

Hence

$$J(0) = (a-g)I + cP. \tag{5}$$

This includes every collision: for $q = 1$ the merged coefficient is $c - 0$; for $q = n-1$ it is $c + 0$; for $q = n/2$ the half-offset coefficient is $0 - 0$. No diagonal collision can revive a vanished term.

Apply P directly to the prescribed Fourier vector:

$$(Pv^{(m)})_j = n^{-1/2} e^{\mathrm{i}\theta_m (j-1)} = e^{-\mathrm{i}\theta_m} v_j^{(m)}.$$

Wraparound is valid since $e^{\mathrm{i}n\theta_m} = 1$. The geometric-series identity shows $(v^{(\ell)})^* v^{(m)} = \delta_{\ell m}$, so the n vectors are an orthonormal basis. Equation (5) gives the full spectrum

$$\begin{aligned} \lambda_m &= a - g + ce^{-\mathrm{i}\theta_m}, \quad 0 \leq m < n, \\ \Re\lambda_m &= a - g + c\cos\theta_m = \mu - c(1-\cos\theta_m), \\ \Im\lambda_m &= -c\sin\theta_m. \end{aligned} \tag{6}$$

Distinct roots of unity remain distinct after multiplying by $c > 0$ and translating. Consequently each of the n eigenvalues has algebraic and geometric multiplicity one over $\mathbb{C}$. Equal real parts of conjugate modes are not repeated eigenvalues. This argument concerns the origin only.

An independent determinant route avoids assuming the spectral formula. Write $t = \lambda - (a - g)$. In the determinant expansion of $tI - cP$, each row can choose its diagonal or predecessor column. If any row chooses its predecessor column, column injectivity forces the preceding row to choose its predecessor too; going around the cycle forces all rows to do so. Thus the only contributing permutations are the identity and the full n-cycle. Their contributions are t^n and $(-1)^{n-1}(-c)^n = -c^n$, respectively. Therefore

$$\chi(\lambda) = (\lambda - (a - g))^n - c^n. \tag{7}$$

At a root, $|\lambda - (a - g)| = c > 0$, so $\chi'(\lambda) = n(\lambda - (a - g))^{n-1} \neq 0$. This independently confirms algebraic simplicity; the already constructed eigenvector gives geometric multiplicity one. P03 **L94–158** passes.

6 Spectral abscissa and exact linear stability

For $0 \leq m < n$, $\cos\theta_m \leq 1$, with equality exactly when $m = 0$. Because $c > 0$,

$$\mathrm{sa}(J(0)) = \mu, \qquad \lambda_0 - \Re\lambda_m = c(1 - \cos\theta_m) > 0 \quad (m \neq 0).$$

The maximizing mode is unique in every odd or even canonical dimension. Neither q nor its gcd with n enters (5); the spectral abscissa and its sign do not even depend on n. Individual nonzero-mode eigenvalues and their zero-real-part loci do depend on n.

Let V have columns $v^{(m)}$ and put $\widehat{z} = V^* z$. Orthonormality and the origin eigenvectors give $\widehat{z}_m(t) = e^{t\lambda_m}\widehat{z}_m(0)$. For real initial data, conjugate Fourier coefficients preserve reality. Thus for $t \geq 0$,

$$\|z(t)\|_2^2 = \sum_m e^{2t\Re\lambda_m}|\widehat{z}_m(0)|^2 \leq e^{2\mu t}\|z(0)\|_2^2.$$

If $\mu < 0$, this proves exponential stability with constant 1 and decay rate $-\mu$. If $\mu > 0$, arbitrarily small real data $\varepsilon\mathbf{1}$ produce $\varepsilon e^{\mu t}\mathbf{1}$ and leave any fixed sufficiently small ball, proving Lyapunov instability.

If $\mu = 0$, the norm is nonincreasing, every nonuniform Fourier component decays, and the uniform component is constant. This proves Lyapunov stability but rules out attraction, asymptotic stability and exponential stability. The three cases exhaust the real parameter μ, proving both directions of all P03 equivalences. In finite dimension these properties do not depend on the choice of norm. No nonlinear stability theorem is used or concluded, especially at $\mu = 0$.

7 Full hyperbolicity-loss set and imaginary pairs

By definition and (6), the exact nonhyperbolicity set in the canonical parameter domain is

$$\bigcup_{m=0}^{n-1} \left\{ (a, k, c, g, h) > 0 : g = a + c\cos\frac{2\pi m}{n} \right\}. \tag{8}$$

Only positive values of the right-hand side are admissible. A proposed locus with $a + c\cos\theta_m \leq 0$ contributes no canonical point at those a, c. Modes m and $n - m$ name the same locus, not two different parameter conditions. Hyperbolicity holds exactly on the complement of this union.

At every nonuniform locus, $\mu = c(1 - \cos\theta_m) > 0$. Thus the origin can be both nonhyperbolic and linearly unstable. The stability boundary is only the $m = 0$ locus; the other components must not be discarded.

For an angular interior representative $0 < m < n/2$, one has $0 < \theta_m < \pi$, hence $\sin\theta_m > 0$. On its admissible locus,

$$\lambda_m = -\mathrm{i}c\sin\theta_m \neq 0, \qquad \lambda_{n-m} = +\mathrm{i}c\sin\theta_m \neq 0.$$

These are distinct simple eigenvalues. On $[0,\pi]$ cosine is strictly decreasing. Choosing representatives $0 \le j \le \lfloor n/2 \rfloor$ therefore allows only one representative at a prescribed zero-real-part value, and exactly the two modes $m, n-m$ are critical for this interior case. All other modes have nonzero real part; this does *not* assert they are all stable. In fact mode zero is already unstable.

For odd n, the highest representative $(n-1)/2$ still satisfies $0 < m < n/2$ and belongs to a pair. For even n, $m = n/2$ is a singleton with zero sine. There is no coincidence of two distinct Fourier pairs at one such locus for fixed $a, c > 0$. This settles both parities. P03 **L168–206** passes. These facts are not a Hopf bifurcation theorem.

8 Zero eigenvalues: boundary and nonuniform singleton

The condition $\lambda_m = 0$ is stronger than $\Re\lambda_m = 0$. Since $c > 0$, its imaginary part forces $\sin\theta_m = 0$. Among canonical mode labels this means $m = 0$, or, only for even n, $m = n/2$.

At the stability boundary $g = a + c$,

$$J(0) = c(P - I), \qquad \lambda_m = c(e^{-\mathrm{i}\theta_m} - 1).$$

Only $m = 0$ has zero real part, and only that mode has eigenvalue zero. Every other real part equals $-c(1-\cos\theta_m) < 0$. The equation $Pz = z$ forces all coordinates equal, so the kernel is $\mathrm{span}\{\mathbf{1}\}$ (over $\mathbb{R}$ or $\mathbb{C}$, respectively). Independently, $\chi(\lambda) = (\lambda + c)^n - c^n$ has $\chi'(0) = nc^{n-1} \neq 0$. Algebraic and geometric multiplicities are both one. For even n, the singleton eigenvalue is $-2c$, not zero.

For a nonzero-mode zero, the only remaining possibility is even n and $m = n/2$, where $\lambda_{n/2} = a - g - c$. Hence its necessary and sufficient condition is

$$n \text{ even}, \quad m = n/2, \quad g = a - c > 0, \quad \text{equivalently } a > c \text{ and } g = a - c.$$

Here $\mu = 2c > 0$, so the synchronous mode is unstable. Also $J(0) = c(I + P)$; $Pz = -z$ gives the one-dimensional alternating kernel $\mathrm{span}\{((-1)^j)_{j=0}^{n-1}\}$. Even parity is necessary for this recurrence to close. The characteristic polynomial is $(\lambda - c)^n - c^n$ and its derivative at zero is $n(-c)^{n-1} = -nc^{n-1} \neq 0$. Thus this zero is algebraically and geometrically simple as well. Odd dimensions have no nonuniform zero. P03 **L208–250** passes without extending its nonlinear scope.

9 Homogeneous coefficients and equilibrium-only simplification

Let $x = r\mathbf{1}$ with $r > -1$ and set $s = 1 + r > 0$. Substitute into the independently derived (3), without assuming an equilibrium:

$$d = a - 2kr - \frac{cr}{s^2} - \frac{g}{s} - hr,$$

$$C = \frac{c}{s}, \qquad G = \frac{gr}{s^2}, \qquad H = hr, \tag{9}$$

$$J(r\mathbf{1}) = dI + CP + GP^{-q} - HP^{q}. \tag{10}$$

The directional action is $dz_i + Cz_{i-1} + Gz_{i+q} - Hz_{i-q}$, which again checks the signs and orientations directly. These formulas hold at arbitrary synchronous states, including negative ones.

Always $C > 0$. For $r > 0$, $G, H > 0$ and the separate neighbor signs are $+, +, -$. For $r = 0$, $G = H = 0$. For $-1 < r < 0$, $G, H < 0$, so the separate neighbor signs are $+, -, +$. At an arbitrary synchronous state there is no universal sign of d; for example at $r = 0$ it is $a - g$. The sign of a merged neighbor entry is decided only after summation.

Only if $r \neq 0$ and $r\mathbf{1}$ is an equilibrium does P02 permit division by r to yield $a-(k+h)r+(c-g)/s = 0$. Substituting $a = (k + h)r - (c - g)/s$ into (9) gives

$$\begin{aligned} d &= (k+h)r - \frac{c-g}{s} - 2kr - \frac{cr}{s^2} - \frac{g}{s} - hr \\ &= -kr - \frac{c}{s} - \frac{cr}{s^2} = -kr - \frac{c(1+2r)}{s^2}. \end{aligned}$$

The h and g cancellations are exact. This works for positive or negative nonzero equilibria. It is not valid at arbitrary synchrony. At the origin the unsimplified value is $a - g$, whereas the displayed simplified expression would be $-c$; they agree only if the additional condition $g = a + c$ is separately imposed. Merely being the origin equilibrium does not supply that condition. P03 makes exactly this restriction at **L295–311**, so no correction is needed.

10 Homogeneous collisions and circulant structure

Use the exhaustive congruence list from Section 4 in (10):

$$\begin{aligned} q = 1: &\qquad J = dI + (C - H)P + GP^{-1}, \\ q = n - 1: &\qquad J = dI + (C + G)P - HP^{-1}, \\ n \text{ even},\ q = n/2: &\qquad J = dI + CP + (G - H)P^{n/2}. \end{aligned}$$

In the second case $P^{-q} = P$ and $P^q = P^{-1}$; in the third, $P^{-q} = P^q = P^{n/2}$. All signs follow from the original directional derivative. The cases are mutually exclusive for $n \geq 5$ and no triple collision occurs. Zero or cancelling coefficients cause no exception.

Since $P^n = I$, the general homogeneous expression is the ordinary polynomial $dI+CP+GP^{n-q}-HP^q$. For every integer r, the entry of P^r in row i, column j is one exactly when $i - j \equiv r$. Thus every entry of the polynomial depends only on $i - j$ modulo n, proving the circulant property at every synchronous state. Linear combination remains valid if powers coincide or their coefficients cancel.

P03's mention of the standard Fourier-diagonalizability consequence does not supply or use the general homogeneous eigenvalue formula. Its full homogeneous spectrum and normality theorem are expressly deferred to P04. Neither is derived or promoted here. P03 **L313–355** passes.

11 Independent synchronous-direction consistency check

Starting directly from P02's scalar field, with a fresh state variable r,

$$f(r) = ar - (k+h)r^2 + \frac{(c-g)r}{1+r}, \qquad f'(r) = a - 2(k+h)r + \frac{c-g}{(1+r)^2}.$$

For $s = 1 + r$, sum the unsimplified coefficients:

$$\begin{aligned} d + C + G - H &= a - 2(k+h)r + \frac{c-g}{s} + \frac{(g-c)r}{s^2} \\ &= a - 2(k+h)r + \frac{(c-g)(s-r)}{s^2} = f'(r). \end{aligned}$$

The identity $s - r = 1$ completes the check. Equivalently differentiate $F(r\mathbf{1}) = f(r)\mathbf{1}$ with respect to r: $J(r\mathbf{1})\mathbf{1} = f'(r)\mathbf{1}$. Every shift fixes $\mathbf{1}$, so collisions preserve the same sum. No equilibrium condition is used. This is a directional/row-sum check, not a derivation of the full positive-equilibrium spectrum. P03 **L375–385** passes.

12 Adversarial exact cases and counterexample search

The universal arguments above establish the verdict. The following exact cases actively test likely failure mechanisms; they are not substitutes for those arguments. Unspecified k, h in origin examples can both be set to 1.

Origin, parity and hyperbolicity stress cases

Take $a = 2, c = 1$. At $g = 3$, every $n = 5, 6, 8$ has one simple zero at $m = 0$ and all other real parts strictly negative. At $g = 4$ all modes decay; at $g = 2$ the synchronous growth rate is 1. At $g = 1$, every even dimension has its simple alternating zero with $\mu = 2$, whereas $n = 5$ has no zero eigenvalue. These examples apply to every canonical q, including collisions.

For interior loci with the same a, c, the following table gives exact critical values. Each listed mode is paired with $n - m$; only that pair has zero real part. Set $\omega = -\Im\lambda_m > 0$.

n	m	g	μ	ω
5	1	$2 + (\sqrt{5} - 1)/4$	$(5 - \sqrt{5})/4$	$\sqrt{10 + 2\sqrt{5}}/4$
5	2	$2 - (\sqrt{5} + 1)/4$	$(5 + \sqrt{5})/4$	$\sqrt{10 - 2\sqrt{5}}/4$
6	1	$5/2$	$1/2$	$\sqrt{3}/2$
6	2	$3/2$	$3/2$	$\sqrt{3}/2$
8	1	$2 + \sqrt{2}/2$	$1 - \sqrt{2}/2$	$\sqrt{2}/2$
8	2	2	1	1
8	3	$2 - \sqrt{2}/2$	$1 + \sqrt{2}/2$	$\sqrt{2}/2$

All these g values are positive. The $n = 5, m = 2$ row specifically checks the largest odd-dimensional representative: it is a nonreal pair, not an even-style singleton. Equal frequencies in different rows do not create simultaneous criticality, since their g values differ.

To test the positivity filter, set $a = 1/4, c = 1, n = 6$. The formal $m = 2$ locus has $g = -1/4$ and is inadmissible; the alternating locus has $g = -3/4$ and is also inadmissible. They cannot be added to the canonical set. Conversely the $m = 1$ locus has $g = 3/4 > 0$ and is retained.

Collisions, gcd, signed states and cancellation

For $n = 5$, $q = 1, 4$ test the two predecessor collisions, while $q = 2, 3$ give distinct neighbors. For $n = 6$, $q = 3$ gives the half-offset collision; $q = 2, 4$ have nontrivial gcd but distinct neighbor offsets. For $n = 8$, $q = 4$ gives the half-offset collision, whereas $q = 2, 6$ have gcd 2 without a collision. Taking $q = 1, n - 1$ in these even dimensions tests the other two cases and their exclusivity.

Two positive nonzero equilibria deliberately force merged coefficients to vanish. At $r = 1$ and $(a, k, c, g, h) = (5/2, 1, 2, 3, 1)$,

$$d = -5/2, \quad C = H = 1, \quad G = 3/4.$$

For $(n,q) = (5,1)$, $J = -\frac{5}{2}I + \frac{3}{4}P^{-1}$: the predecessor contribution cancels completely. At $r = 1$ and $(a,k,c,g,h) = (7/2,1,1,4,1)$,

$$d = -7/4, \quad C = 1/2, \quad G = H = 1.$$

For $(n,q) = (6,3)$ or $(8,4)$, $J = -\frac{7}{4}I + \frac{1}{2}P$. Each tuple satisfies the equilibrium relation before it is used to simplify d. These examples refute any unstated requirement that every listed cyclic diagonal be nonzero, but they do not refute P03.

Cancellation of $C + G$ is possible at negative synchrony as well: take $(n,q) = (6,5)$, $r = -1/2$ and $(a,k,c,g,h) = (4,1/2,2,2,1)$. Then $(d,C,G,H) = (5,4,-4,-1/2)$ and $J = 5I + \frac{1}{2}P^{-1}$. This is an arbitrary synchronous state, not an asserted equilibrium. P03's additive formulas cover it; its two named cancellation types are examples, not a purported exhaustive classification of cancellations.

A negative nonzero equilibrium in the open domain is $r = -1/2$ with $(a,k,c,g,h) = (1,1,1,2,1)$. It has

$$d = 1/2, \quad C = 2, \quad G = -4, \quad H = -1/2, \quad f'(r) = -1.$$

Indeed $a-(k+h)r+(c-g)/(1+r) = 1+1-2 = 0$ and $d+C+G-H = -1$. The three possible merged coefficients are respectively $C - H = 5/2$, $C + G = -2$, and $G - H = -7/2$. This tests negative-state sign reversals and confirms that d need not be negative even at a negative equilibrium. For comparison, the same r with all five parameters equal to 1 is not an equilibrium: its unsimplified $d = 5/2$, while the equilibrium-only formula would give $1/2$. At the origin with all five parameters equal to 1, the correct $d = 0$, while that formula would give -1. These are adversarial controls against misuse, not errors in P03.

Reproducible finite corroboration

The accompanying exact checker differentiates the original five-term field using rational automatic differentiation, with actual modular indices identified first. It compares the result against the independently written entry and shift expressions for all q in dimensions $5, 6, 8$, including nonsynchronous signed rational states, positive and negative synchrony, the origin, nonzero equilibria and cancellations. All arithmetic in that checker is exact; its script and machine-readable results are included in the audit-only package. The results report actual counts and checks. The hand proofs remain necessary for arbitrary n, q, x and parameters.

The independent run passes 656 full Jacobian cases, comprising 29,820 exact entry comparisons over all 16 dimension/offset pairs. It includes 456 homogeneous cases (20,734 matrix entries), 3,020 scalar row-sum comparisons, 48 origin cases, and 112 actual nonzero equilibria (64 positive and 48 negative). At those equilibria 742 original field components vanish and 742 diagonal simplifications agree exactly. All three merged-coefficient cancellation types are covered. Counts for overlapping identities are not added as if they were independent samples. Separate exact sparse-determinant and rational-rank checks at $n = 5, 6, 8$ corroborate the origin calculation.

The earlier P03 computational paragraph is explicitly corroborative and is not imported as evidence of proof. Its claimed sample counts are arithmetically consistent:

$$\sum_{n=5}^{10}(n-1) = 39, \quad \sum_{n=5}^{10}(n-1)n^2 = 2570, \quad 8 \cdot 39 = 312, \quad 8 \cdot 2570 = 20560.$$

Historical residuals are properties of their recorded program and environment, not universal mathematical error bounds. The independent checking record distinguishes those reports from the exact checks performed for A03.

The matching original companion was located at `output/p03/P03_validation_support.zip`. Its checker has the manifest's SHA-256

`73784fd269f445a80639175b8901610e89a517469d69ddc5d397200dc5412bbb`

The inspected checker was copied and rerun without modifying the original reports or reference files. In Python 3.12.14 and NumPy 2.3.5, its exact and floating result records match the originals, including the maximum scaled finite-difference error $2.6907607135803086 \times 10^{-9}$, correctly rounded up to 2.70×10^{-9} in P03. The maximum absolute difference is $7.230752174791633 \times 10^{-8}$; the origin eigenvector residual is $1.1889678785578623 \times 10^{-15}$. The reproduction comparison and summary are retained with A03. This confirms the reported finite experiment, not the universally quantified theorem statements.

No counterexample to a mathematical P03 claim was found. More importantly, the derivations above explain why the attempted failures cannot occur under the canonical hypotheses.

13 Terminology, scope and dependency verification

P03 uses "asymptotically", "exponentially", "Lyapunov stable" and "unstable" in **L163–193** for the explicitly specified linear system. It does not claim that nonlinear asymptotic stability is equivalent to $\mu < 0$. At $\mu = 0$, it classifies only the linear system. "Nonhyperbolic" in **L168–176,244–250** is correctly based on zero real part, and is allowed to overlap linear instability. The interior imaginary eigenvalues in **L198–206** are nonzero. The zero-eigenvalue results in **L208–242** impose the additional zero-imaginary-part requirement. No conflation is present.

The only dependencies used are the canonical field and conventions, P01's smoothness on the open domain, P02's scalar restriction and explicitly invoked nonzero-equilibrium relation, elementary differentiation, elementary cyclic permutation/Fourier facts, and scalar linear differential equations. P01's uniqueness supports P02's synchronous restriction but is not an extra spectral premise. P01's global absorbing-box conclusions and P02's full root classification are not needed by P03's linear algebra.

Current P01 v1.1 has full-file hash

`0d6950821aa83f3d09e8bfe84dab2b0b5020aa6e7881cf6928e984bff1890f1e`

and current P02 v1.1 has full-file hash

`8e057c2cd009659eee45906844e25f4f38c156e5eb2e4130e824aa5e8299c51a`

Their current corrections are present: P01's larger-real-root wording and P02's explicitly ordered (s_-, s_+) boundary limits. A01/A02 retain their original verdicts on earlier exact versions; the later closure records, current headers and manifest establish the accepted prerequisite state. Restored A02 source is present despite P02's preserved historical availability comment. Historical all-TODO and DRAFT sentences are not treated as current status instructions.

P03 does not use a P04 theorem to prove its origin result. Its origin Fourier calculation is supplied in full. The old notation document's broad ownership sentence about conjugacy/simplicity must be read with the current P03 origin scope, which is explicit in the current manifest, registry and compendium. This is historical scope bookkeeping, not a circular mathematical dependency.

No full positive-equilibrium Fourier spectrum, homogeneous normality theorem, positive-state mode selection, nonlinear boundary stability, transcritical bifurcation, Hopf bifurcation, first Lyapunov coefficient or rotating wave is proved or claimed complete in P03. Origin spectral ordering is part of its authorized scope. P03's explicit deferrals at **L22–25,244–250,320–322** are respected. No new literature search was conducted, and the frozen priority limitation remains unchanged.

14 Line-by-line issue and disposition list

Required issue list: empty. The following contiguous line ranges cover every line of the exact 433-line source, including nonmathematical headers, comments and whitespace. "Pass" is a mathematical/scope finding; "Record" identifies status or computational provenance rather than a theorem.

P03 lines	Disposition	Independent finding
1–11	Record	Correct v1.0 DRAFT provenance, prerequisite and origin-only scope labels. No numerical dependency asserted.
12–26	Pass	Canonical assumptions, domain and abbreviations exact; linear/nonlinear distinction and P04/P09 deferrals explicit.
27–32	Pass	Modular indicators correctly defined as additive.
33–48	Pass	General entry formula correct; neighbors off diagonal but potentially mutually coincident.
49–59	Pass	Both rational derivatives correct, including opposite denominator-derivative signs.
60–71	Pass	Loss-denominator derivative has sign of x_i; polynomial derivatives and additive handling correct.
72–79	Pass	D_i, C_i, G_i, H_i match the separately differentiated terms.
80–90	Pass	Directional formula and P, P^{-q}, P^q orientations exact.
91–93	Pass	Left diagonal factors use receiving-row coefficients; no general-state circulant claim.
94–104	Pass	Origin equilibrium and every vanishing/surviving derivative identified correctly.
105–109	Pass	$J(0) = (a-g)I + cP$, including collisions.
110–118	Pass	Complete origin spectrum with negative exponent; all eigenvalues simple over $\mathbb{C}$.
119–130	Pass	Real/imaginary parts, unique maximizing mode and spectral abscissa correct.
131–142	Pass	Componentwise shift, wraparound and geometric-sum basis proof correct.
143–152	Pass	Distinctness, characteristic polynomial and nonzero derivative prove multiplicities.
153–159	Pass	Strict cosine inequality for all nonzero modes, all parities.
160–177	Pass	Linear stability trichotomy and full nonhyperbolicity set correct; $g > 0$ retained.
178–193	Pass	Fourier solution, norm bounds, real growing solution and both converses correct.
194–200	Pass	Independence from q/gcd and threshold independence from n correct; $n = 6$ example exact and no Hopf assertion.
201–207	Pass	Exactly one angular interior conjugate pair can be critical. For odd n, the largest representative is still interior in $0 < m < n/2$.
208–214	Pass	Boundary zero unique, simple, with synchronous kernel; other modes strictly left.
215–229	Pass	Direct kernel, determinant derivative and even singleton $-2c$ establish every claim.
230–242	Pass	Nonuniform zero iff even n, $m = n/2$, $g = a - c > 0$; simple, with synchronous instability.
243–251	Pass	Nonhyperbolic/unstable overlap and linear/nonlinear distinction correctly stated; no transcritical claim.
252–257	Pass	All synchronous states $x^* > -1$ included; local use of $s = 1 + x^*$ disclosed.
258–276	Pass	Unsimplified homogeneous coefficients and summed entries exact without equilibrium assumptions.
277–288	Pass	Direct substitution and index-based shift proof valid.
289–294	Pass	Positive, zero and negative-state sign qualifications correct; signs after collisions not assumed.
295–301	Pass	P02 relation used only for a nonzero synchronous equilibrium.

P03 lines	Disposition	Independent finding
302–312	Pass	Both simplified forms of d exact; arbitrary states and automatic use at origin explicitly excluded.
313–323	Pass	Polynomial in P proves circulant structure; generic spectrum and normality deferred.
324–335	Pass	Pairwise congruence list exhaustive; diagonal/triple collisions excluded; no extra gcd case.
336–348	Pass	Three homogeneous collision forms, exclusivity and possible zero coefficients correct.
349–356	Pass	General-state merged entries use the identified coordinate and add all terms; cancellation allowed.
357–374	Pass	First-order inverse expansion and independent origin substitution reproduce derivative.
375–386	Pass	Scalar derivative equals row sum without the equilibrium relation.
387–393	Pass	Independent multiplicity/order checks valid; equality of real parts distinguished from eigenvalue multiplicity.
394–409	Record	Finite corroboration explicitly not proof. Counts are consistent; reproduced checks and provenance are recorded separately. No later spectrum or nonlinear result is inferred.
410–432	Record	Draft auditor checklist is reference material. All relevant requested targets are addressed by this independently commissioned audit.
433	Record	Historical DRAFT-status statement remains unchanged; A03 supplies the new independent verdict.

One optional clarity note, not a required revision: at **L201–206**, "interior representative" is most explicitly read as $0 < m < n/2$. For odd n, this includes the largest representative $\lfloor n/2 \rfloor$. The full all-mode formula and the project's singleton convention already give the correct result. This reading is verified explicitly above and does not change the verdict.

15 Downstream consequences and closure

The exact P03 v1.0 identified in Section 1 has passed the independent mathematical audit. There is no unresolved correction preventing closure. The accepted outputs are the general Jacobian with additive collisions, the complete origin spectrum and linear classification, the full origin nonhyperbolicity/zero-eigenvalue analysis, and the unsimplified homogeneous circulant representation with its correctly restricted simplification.

P04 is cleared to begin after A03 closure is recorded against this exact P03 hash. Closure can cite this audit and its VERIFIED verdict while preserving the mathematical P03 source. The manifest/registry/compendium have not been changed in this audit-only delivery. If a future administrative header change alters P03's full-file hash, that closure record should preserve the audited hash and explicitly account for the metadata-only difference. A substantive proof change requires review of the changed version.

Clearance is permission to attempt P04, not verification of P04 or of a positive-equilibrium spectrum, normality, bifurcation or later-module claim. P09 still has its other dependencies; this audit does not decide nonlinear behavior at $\mu = 0$. The frozen literature-priority limitation remains a separate publication gate. No P04 work or new literature work is performed.

The audit-only package contains the TeX source, PDF reading copy, input fingerprints, reproducible checks and provenance. Every supplied project member is preserved byte-for-byte.